Important update: Arm Announces End of Life Timeline for Mbed. This site will be archived in July 2026. **Read the full announcement.** (https://os.mbed.com/blog/entry/Important-Update-on-Mbed/)

To compile a program for this board using Mbed CLI, use **NUCLEO_G071RB** as the target name.

NUCLEO-G071RB

Affordable and flexible platform to ease prototyping using a STM32G071RB microcontroller.

Table of Contents

1. Overview
2. Microcontroller features
3. Nucleo features
4. Nucleo pinout
5. Supported shields
6. Getting started
7. Technical references
8. Example applications
9. Known limitations
10. Tips and Tricks

Overview

The STM32 Nucleo-64 boards provide an affordable and flexible way for users to try out new concepts and build prototypes by choosing from the various combinations of performance and power consumption features, provided by the STM32 microcontroller.

The Arduino™ Uno V3 connectivity support and the ST morpho headers allow the easy expansion of the functionality of the STM32 Nucleo open development platform with a wide choice of specialized shields.

The STM32 Nucleo-64 board does not require any separate probe as it integrates the ST-LINK debugger/programmer.

This board is not yet fully Mbed enabled, so not available with online compiler. It is available only with Mbed-CLI and on GitHub repository.

Microcontroller features

- Core: Arm® 32-bit Cortex®-M0+ CPU, frequency up to 64 MHz
- -40°C to 85°C/125°C operating temperature
- Memories
  - Up to 128 Kbytes of Flash memory
  - 36 Kbytes of SRAM (32 Kbytes with HW parity check)
- CRC calculation unit
- Reset and power management
  - Voltage range: 1.7 V to 3.6 V
  - Power-on/Power-down reset (POR/PDR)
  - Programmable Brownout reset (BOR)
  - Programmable voltage detector (PVD)
  - Low-power modes:Sleep, Stop, Standby, Shutdown
  - VBAT supply for RTC and backup registers
- Clock management
  - 4 to 48 MHz crystal oscillator
  - 32 kHz crystal oscillator with calibration
  - Internal 16 MHz RC with PLL option (±1 %)
  - Internal 32 kHz RC oscillator (±5 %)
- Up to 60 fast I/Os
  - All mappable on external interrupt vectors
  - Multiple 5 V-tolerant I/Os
- 7-channel DMA controller with flexible mapping
- 12-bit, 0.4 µs ADC (up to 16 ext. channels)
  - Up to 16-bit with hardware oversampling
  - Conversion range: 0 to 3.6V
- Two 12-bit DACs, low-power sample-and-hold
- Two fast low-power analog comparators, with programmable input and output, rail-to-rail

Board Partner

(/teams/ST/)

ST (/teams/ST/)

A world leader in providing the semiconductor solutions that make a positive contribution to people's lives, both today and in the future.

Buy Now (https://www.st.com/content/st_com/en/products/evaluation-tools/product-evaluation-tools/mcu-mpu-eval-tools/stm32-mcu-mpu-eval-tools/stm32-nucleo-boards/nucleo-g071rb.html)

Mbed Enabled

- Baseline

Mbed OS support

- Mbed OS 6.10
- Mbed OS 6.11
- Mbed OS 6.12
- Mbed OS 6.13
- Mbed OS 6.14
- Mbed OS 6.15

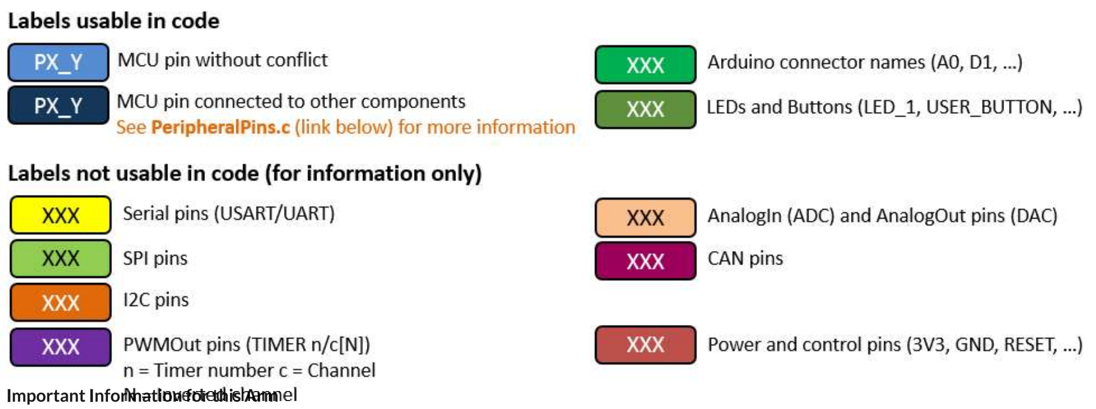
- 14 timers (two 128 MHz capable): 16-bit for advanced motor control, one 32-bit and five 16-bit general-purpose, two basic 16-bit, two low-power 16-bit, two watchdogs, SysTick timer
- Calendar RTC with alarm and periodic wakeup from Stop/Standby/Shutdown
- Communication interfaces
  - Two I2C-bus interfaces supporting Fast-mode Plus (1 Mbit/s) with extra current sink, one supporting SMBus/PMBus and wakeup from Stop mode
  - Four USARTs with master/slave synchronous SPI; two supporting ISO7816 interface, LIN, IrDA capability, auto baud rate detection and wakeup feature
  - Low-power UART
  - Two SPIs (32 Mbit/s) with 4- to 16-bit programmable bitframe, one multiplexed with I2S interface
  - HDMI CEC interface, wakeup on header reception
- USB Type-C™ Power Delivery controller
- Development support: serial wire debug (SWD)
- 96-bit unique ID
- All packages ECOPACK®2 compliant

Nucleo features

- Common features
  - STM32 microcontroller in LQFP64 package
  - 1 user LED shared with Arduino™
  - 1 user and 1 reset push-buttons
  - 32.768 kHz crystal oscillator
  - Board connectors:Arduino™ Uno V3 expansion connectorST morpho extension pin headers for full access to all STM32 I/Os
  - Flexible power-supply options: ST-LINK, USB VBUS or external sources
  - On-board ST-LINK debugger/programmer with USB re-enumeration capability: mass storage, Virtual COM port and debug port
  - Comprehensive free software libraries and examples available with the STM32Cube MCU Package
  - Support of a wide choice of Integrated Development Environments (IDEs) including IAR™, Keil® and GCC-based IDEs
- Board-specific features
  - External SMPS to generate Vcore logic supply
  - 24 MHz HSE
  - Board connectors:External SMPS experimentation dedicated connectorMicro-AB or Mini-AB USB connector for the ST-LINKMIPI® debug connector

Nucleo pinout

Pins Legend



You can find more details on the available pins and labels in the **PeripheralPins.c** and **PinNames.h** files.

These files can be found in:

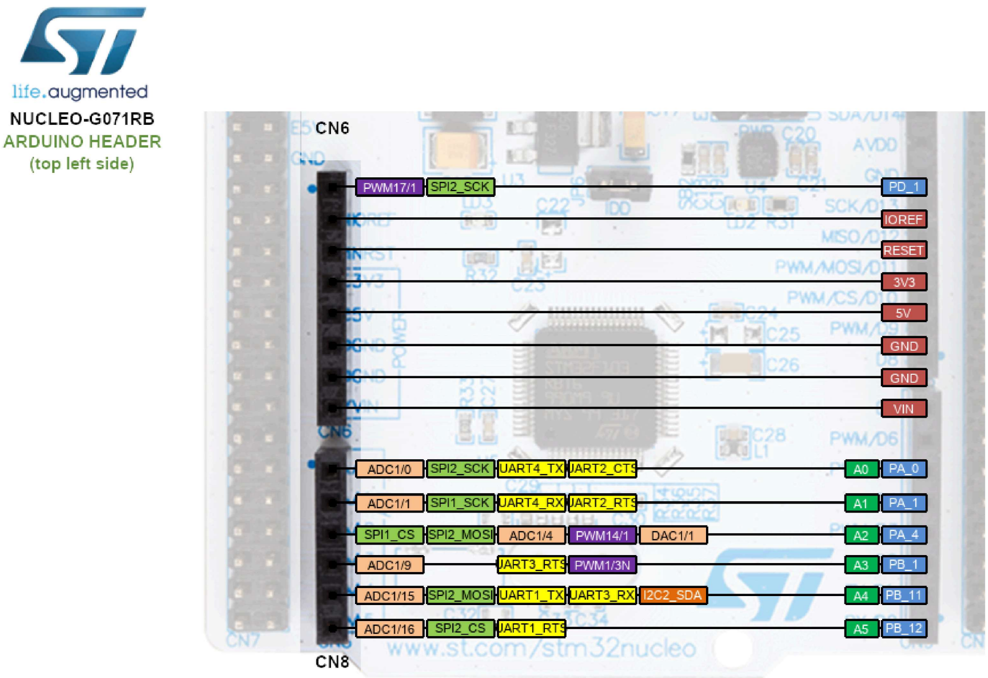
- ARMmbed/mbed-os repository on GitHub (**up-to-date** version, used with **mbed CLI commands**)

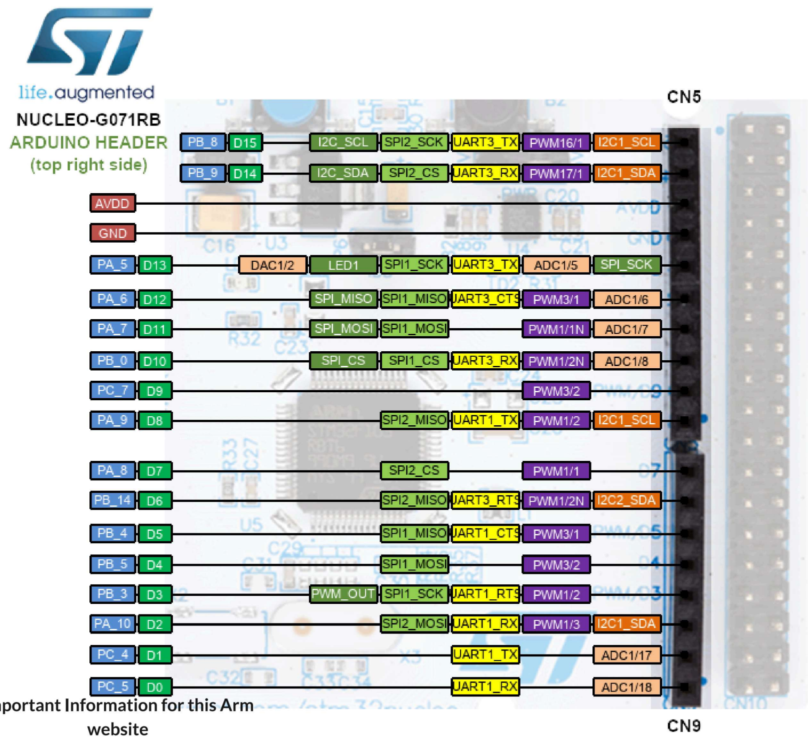
https://github.com/ARMmbed/mbed-os/blob/master/targets/TARGET_STM/TARGET_STM32G0/TARGET_STM32G071xx/TARGET_NUCLEO_G071RB/ (https://github.com/ARMmbed/mbed-os/blob/master/targets/TARGET_STM/TARGET_STM32G0/TARGET_STM32G071xx/TARGET_NUCLEO_G071RB/)

- mbed-dev library in developer.mbed.org (source files of the mbed library used on **mbed compiler IDE**)

https://developer.mbed.org/users/mbed_official/code/mbed-dev/file/default/targets/TARGET_STM/TARGET_STM32G0/TARGET_STM32G071xx/TARGET_NUCLEO_G071RB/ (https://developer.mbed.org/users/mbed_official/code/mbed-dev/file/default/targets/TARGET_STM/TARGET_STM32G0/TARGET_STM32G071xx/TARGET_NUCLEO_G071RB/)

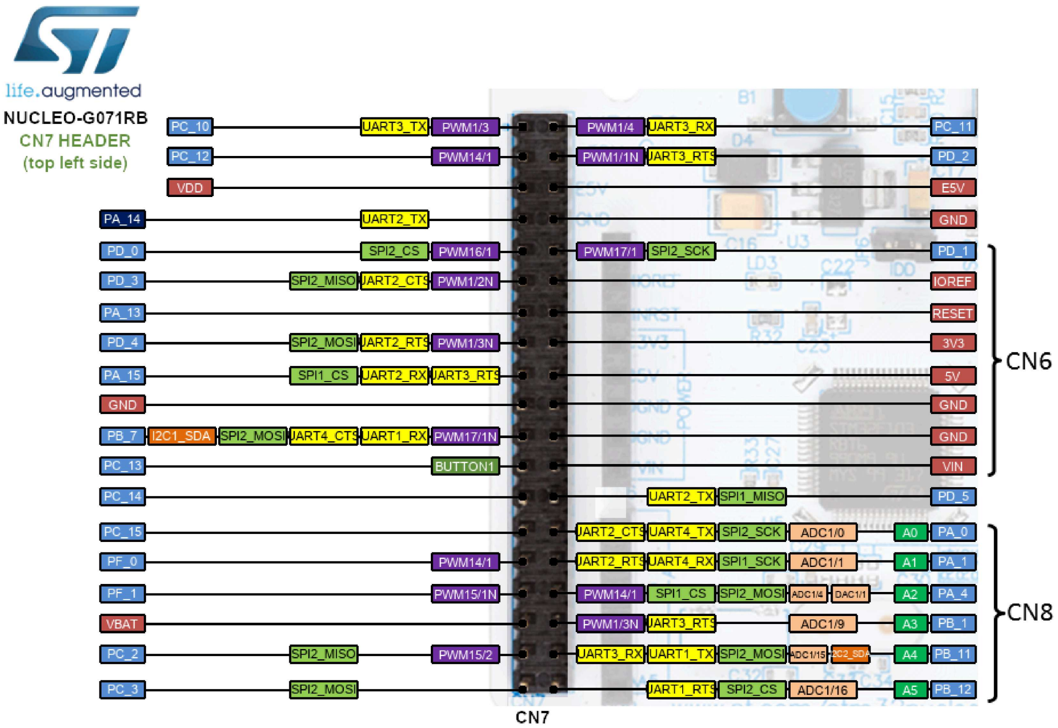
Arduino-compatible headers

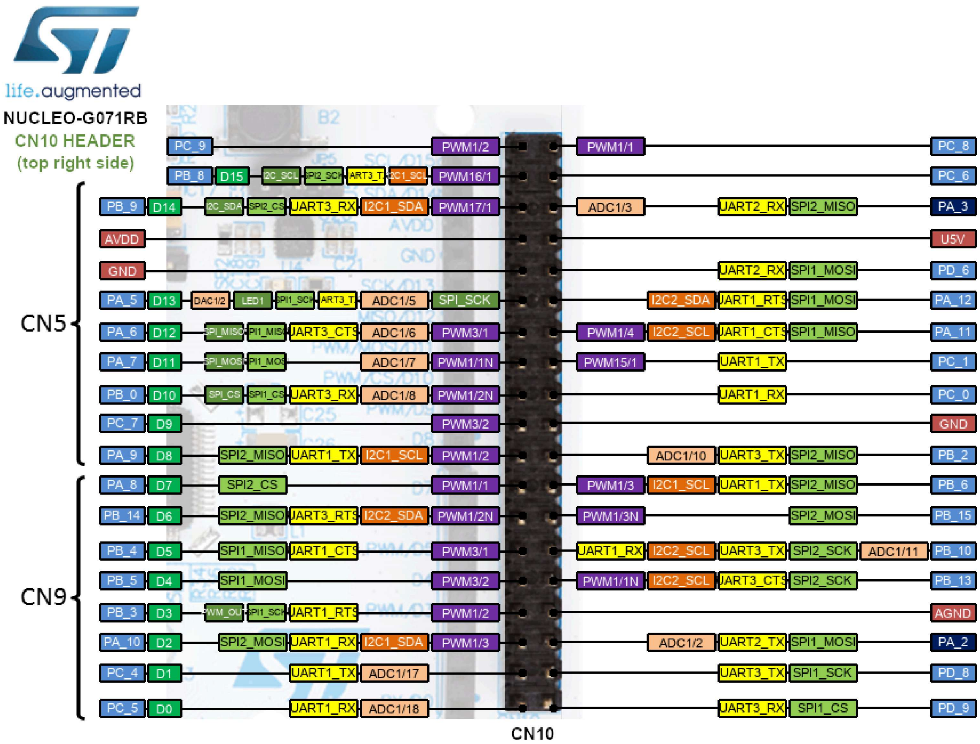




Morpho headers

These headers give access to all STM32 pins.





Supported shields

ST X-NUCLEO boards

See Matrix of tested boards (https://developer.mbed.org/teams/ST/wiki/Matrix-of-tested-boards).

Other Non-ST boards

See here (https://developer.mbed.org/teams/ST/wiki/Supported-shields).

Getting started

Nucleo ST-LINK/V2 driver installation and firmware upgrade

- Install the ST-LINK/V2 driver before connecting the Nucleo board to your PC the first time. Follow this **LINK (/teams/ST/wiki/ST-Link-Driver)** for all details.
- For optimum performances, ensure that the Nucleo ST-LINK/V2 firmware is upgraded to the latest version. Follow this **LINK (/teams/ST/wiki/Nucleo-Firmware)** for all details.

Technical references

For more information, please refer to:

- MCU STM32G071RB (https://www.st.com/content/st_com/en/products/microcontrollers-microprocessors/stm32-32-bit-arm-cortex-mcus/stm32-mainstream-mcus/stm32g0-series/stm32g0x1/stm32g071rb.html)
- Nucleo boards (http://www.st.com/stm32nucleo)
- Nucleo-G071RB development board (https://www.st.com/content/st_com/en/products/evaluation-tools/product-evaluation-tools/mcu-mpu-eval-tools/stm32-mcu-mpu-eval-tools/stm32-nucleo-boards/nucleo-g071rb.html)

Example applications

- mbed-os-example-blinky-baremetal (https://os.mbed.com/teams/mbed-os-examples/code/mbed-os-example-blinky-baremetal)

Known limitations

The following section describes known limitations of the platform. Note that general issues are tracked into the mbed repository (https://github.com/mbedmicro/mbed) available on GitHub.

- On Nucleo 64-pins boards, the D0 and D1 pins are not available per default as they are used by the STLink Virtual Comm Port. More information HERE (https://os.mbed.com/teams/ST/wiki/Use-of-D0D1-Arduino-pins)

Tips and Tricks

Find more information in ST WIKI pages (https://os.mbed.com/teams/ST/wiki/Special:Allpages)..

Buy Now (https://www.st.com/content/st_com/en/products/evaluation-tools/product-evaluation-tools/mcu-mpu-eval-tools/stm32-mcu-mpu-eval-tools/stm32-nucleo-boards/nucleo-g071rb.html)

You need to log in (/account/login/?next=/platforms/ST-Nucleo-G071RB/) to post a discussion